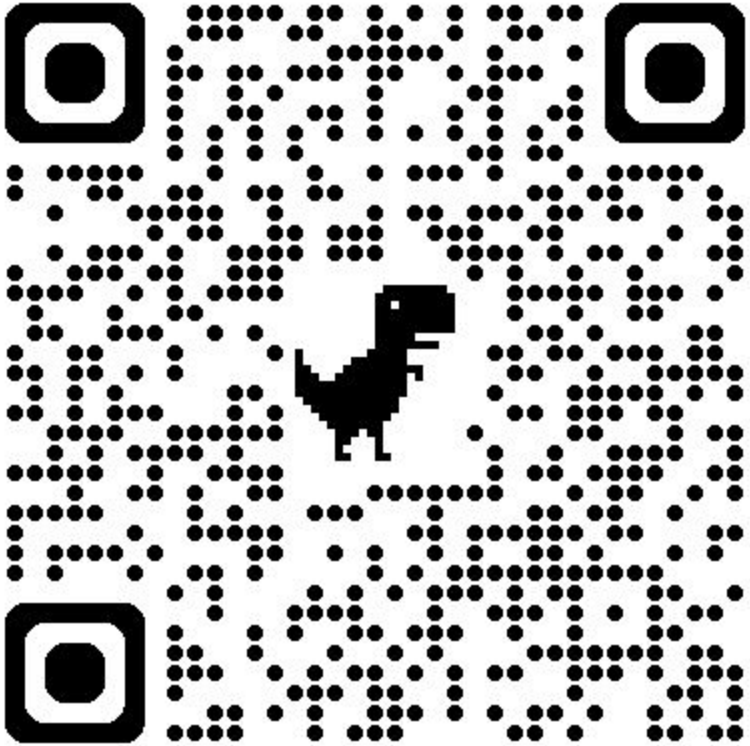
Scan the Dino to access all slides for CS010A Labs

- Kudos to Vedant Saqcena for coming up with this catchphrase :)



Week 4 Lab - CS010A

Matching Digits

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Problem Overview

1. You are given a number between 11 and 100.
2. You have to find the largest number less than (or equal to) the given number that has matching digits.
    a. Actual problem needs some more work, we'll get there, but this is the overall idea.

Example:

- For 90, 88 is the largest number that has matching digits.
- For 54, 44 is the largest number that has matching digits.
- For 77, 77 is the largest number that has matching digits.

Problem Details

- Input: 1 integer inputs
  - Range from 11-100, inclusive
  - Assume input >0, no need to check for this
- Output:
  - 1. **Countdown** that begins at your chosen integer, stopping when the 2 digits in your integer are identical
    - Specifics:
      - Print each number in the countdown followed by a space
      - Once matching digits found, end the countdown with newline
  - 2. Print **distance** from original input to the final
- Review:
  - 1 Input: 1 integer
  - 2 Outputs: A countdown, and the distance

Dry run with an example test case

Input:

35

Expected output:

35 34 33

2

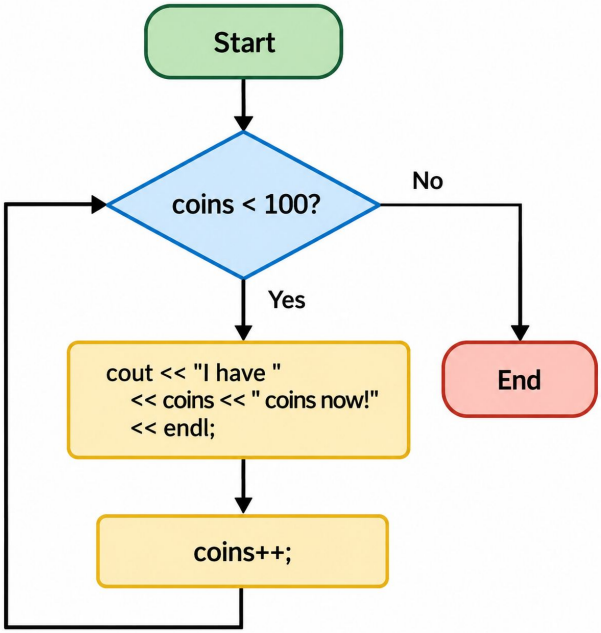
Input:

5

Expected output:

Input must be 11-100

Loop basics: while loop



Syntax:

```
while (condition) {
      // Repeated statements
}
```

Condition: Similar to condition in if-else statements

```
( coins < 100 )
```

Block contents: Set of statements we want to execute at each step.

```
cout << "I have " << coins <<  " coins now!" << endl;
```

Variant: Statement that changes the variable inside the condition

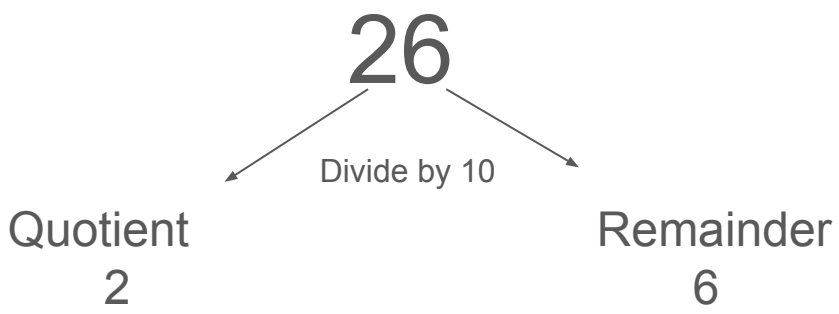
```
coins++;
```

Finding digits

```
int number = 26;
```

tens_digit = 2

units_digit = 6




```
tens_digit  = number / 10;
units_digit = number % 10;
```

Code plan tips - Be specific!

- Input (what variable are you taking in?)
- Are we checking for something? Yes!
  - We need to check that the input is within 11 - 100
- While loop
  - 3 components (condition, body, what gets changed)
  - Example: While this condition is not satisfied, we print this to user, and change this variable by so much
- input / 10
  - This divides by 10 and leaves you with the tens digit
- input % 10
  - This is the remainder after dividing by 10, which is your ones digit
- Output (what do you expect to get in your output? Describe it!)

Hard requirements

- Do’s
  - Must use a while loop
  - Must compare digits
- Do not’s
  - Do not use “true” as the condition in your while loops
    - Ex: while (true) { }
  - Do not us “break;” in your while loops
  - Do not hardcode if-else statements for each multiple of 11